Deokhyun *Ahn*

Ph.D. Candidate · Brain & Cognitive Engineering

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01 RESEARCH INTERESTS

Video understanding, contextual dynamics, and efficient AI systems for social good.

My research focuses on video understanding, contextual dynamics, and efficient AI systems for social good. Starting from sensing-based applications, including a SLAM-based smart helmet for firefighters and indoor drone trajectory tracking, I became interested in reliable real-world systems. I now study how video models can move beyond RGB and motion cues to capture contextual interactions, and recently I have focused on VLM-guided video anomaly detection for surveillance, with emphasis on target-domain transfer and efficient sampling.

- VIDEO INTELLIGENCE
- VIDEO UNDERSTANDING
- VIDEO ANOMALY DETECTION
- VIDEO ACTION RECOGNITION

02 EDUCATION

Mar 2023 – Present
Seoul, KR

Ph.D., Brain & Cognitive Engineering

Korea University

- Video anomaly detection with a focus on modeling human-inspired contextual understanding in video.

Aug 2015 – Feb 2020
Seoul, KR

M.Eng., Electronic Engineering

Kwangwoon University

- Thesis — *A Study on Passive Target Tracking in Three-Dimensional Indoor Environment by Using Deep Neural Network*.
- Completed 19 months of mandatory military service.

Mar 2013 – Aug 2015
Seoul, KR

B.Eng., Electronic Engineering TRANSFER

Kwangwoon University

- Transferred in Mar 2013 from National Institute for Lifelong Education, Ministry of Education — B.Eng. in Computer Engineering coursework, Mar 2011 – Feb 2013.

03 RESEARCH & WORK EXPERIENCE

Jan 2023 – Present
Seoul, KR

Student Researcher CURRENT

Korea Institute of Science and Technology (KIST)

Video anomaly understanding · Surveillance video · DNN

- Modeled contextual dynamics by integrating RGB appearance, motion dynamics, and scene information — enabling the model to better capture anomaly concepts through their relationships. [1] [2] [3] [4]

Jan 2022 – Feb 2023
Seoul, KR

Research Intern

Korea Institute of Science and Technology (KIST)

Video anomaly understanding in surveillance videos & DNN

Oct 2021 – Dec 2021
Sejong, KR

Researcher

The Korea Transport Institute

Autonomous bus development · Project coordination

04 RESEARCH GRANTS & PROJECTS

~$24M
5-year program

Real-Time Intelligent Surveillance for Public Safety NATIONAL R&D

MSIT & MOTIE · with Yonsei University and the Korean National Police Agency

Context-aware video anomaly detection across city-scale CCTV control centers (~1,000 cameras); field-validated in Anyang City.

~$9.5M
5-year program

Mobile Platform for Active Safety Services

Korean National Police Agency & MOTIE

Government R&D · mobile-platform-based solution for proactive public-safety services.

New
initiative

Anomaly Detection in Correctional Facilities

KIST AI·Robotics

Detecting inmate anomalous behavior in custodial environments.

05 SELECTED PUBLICATIONS

[1] **Ahn D.**, Kim H., Han J.-H., Ham B., Choi H., Kim I.-J., Kim H. ACCEPTED
Look Closer: Action-Guided Dense Visual Dynamics for Proficiency Estimation.
CVPR 2026 Workshop on Multimodal Human Motion Analysis · Pitch & Poster

[2] **Ahn D.**, Jo Y., Lee U.-S., Park H., Han J.-H., Kim H. UNDER REVIEW
Learning Robust Representations for Few-Shot Action Recognition with Frame-Level Ambiguities.
BMVC 2026

[3] **Ahn D.**, Choi Y., Han J.-H., Kim I.-J., Kim H. UNDER REVIEW
BUSTER: Adaptive Sampling for VLM-Guided Unsupervised Video Anomaly Detection.
BMVC 2026

[4] **Ahn D.**, Jo Y., Kim D., Nam G. P., Han J.-H., Kim H. PUBLISHED
Where and What: Contextual Dynamics-Aware Anomaly Detection in Surveillance Videos.
IEEE Transactions on Image Processing, 2025 · IF 13.7 · top 2.2% in E.E. Eng.

[5] Kim D., **Ahn D.**, Jo Y., Park H., Lee S., Kim H. PUBLISHED
A Foundational Research Framework for Real-World Abandoned Object Detection: Train-Free Baseline and a Standardized Benchmark.
Expert Systems with Applications, 2025 · 130658

[6] Zhou B., **Ahn D.**, Lee J., Sun C., Ahmed S., Kim Y. PUBLISHED
A Passive Tracking System Based on Geometric Constraints in Adaptive Wireless Sensor Networks.
Sensors, 18(10), 3276, 2018

[7] **Ahn D.**, Kim N. M., Park J. H., Kim Y. O. PUBLISHED
2D Indoor Map Building Scheme Using Ultrasonic Module.
J. Korean Inst. Comm. and Info. Sci., 41(8), 986–994, 2016

06 AWARDS & HONORS

Dec 2025

AIR Outstanding Research Award

KIST AI·Robotics Institute

> *In recognition of outstanding research achievements during 2025.*

Feb 2025

Excellent Paper Award – Graduate School

Korea University

> *For outstanding research performance and publication of an excellent paper during doctoral studies.*

Nov 2024

Outstanding Oral Presentation Award

KIST Academia-Research Convergence Conference

> *For the presentation "Multi-Aspect Contextual Dynamics for Anomaly Detection in Surveillance Videos."*

Jul 2021

Division Commander's Commendation

Capital Mechanized Infantry Division

> *For a combat development proposal on reconnaissance drone AI improvement and AI/big-data-based acoustic detection.*

07 TEACHING

Sep – Dec 2018

Senior Capstone Project

Kwangwoon University · Instructor: Prof. Youngok Kim

- Assisted in managing and supporting the Capstone Design Project as a TA for 50+ students.

Mar – Jun 2017

Computer Architecture

Kwangwoon University · Instructor: Prof. Youngok Kim

- Graded assignments and provided feedback for 30+ students with Q&A sessions.

08 TECHNICAL SKILLS

Programming

Python · C · Java · Git · Linux

Machine Learning

PyTorch · TensorFlow · Keras · NumPy · Pandas · Matplotlib

Sensors

Ultrasonic · Beacon · RF · Wi-Fi · 3-axis Accelerometer

Embedded & IoT

Raspberry Pi · Arduino